Self-Organized Criticality

The Bak–Tang–Wiesenfeld Sandpile Model

Power Laws, Avalanches, and Criticality Without Tuning

From Conway to LangGraph • Week 4

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Today's Roadmap

I **From Tuned to Self-Organized Criticality**
Why so many systems live on the edge

II **The BTW Sandpile Model**
Rules, avalanches, and the critical attractor

III **Power Laws and Scaling**
Heavy tails, exponents, and how (not) to fit them

IV **Implementation in NumPy**
Vectorized sandpile simulation

V **Analysis and Exercises**
Avalanche statistics, finite-size effects, universality

PART I

From Tuned to Self-Organized Criticality

Why so many systems live on the edge

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Last Week: Criticality Requires Tuning

Ising Model

Critical behavior (divergent correlation length, power-law correlations) occurs only at $T = T_c$.

You must fine-tune the temperature to see criticality.

Above or below T_c → exponential decay, finite correlations.

Schelling Model

Sharp phase transition at $T_c \approx 0.30$.

Maximum fluctuations and "susceptibility" peak at T_c.

Away from T_c the system is either well-mixed or fully segregated — boring.

The Problem

In nature, we observe power laws everywhere: earthquake magnitudes, forest fire sizes, neural avalanches, solar flares, extinction events. These systems don't have an external "knob" set to a critical value. So how do they reach criticality?

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Self-Organized Criticality (SOC)

Bak, Tang & Wiesenfeld (1987)

Core idea: Some driven dissipative systems naturally evolve toward a critical state — without any parameter tuning. The critical state is an **attractor** of the dynamics, not a special parameter value.

Slow driving

Energy/mass added to the system gradually (one grain at a time)

Threshold dynamics

Local instabilities trigger when a threshold is exceeded

Dissipation at boundaries

Energy leaves the system at the edges (open boundaries)

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The Sand Pile Analogy

Imagine slowly dropping sand grains onto a table, one at a time.

Early — **Subcritical**

Early on, grains just pile up. The pile grows steeper. Nothing interesting happens — small, local rearrangements.

Steady state — **Critical state**

Eventually the slope reaches a critical angle. Now each new grain may trigger an avalanche of any size — from a single grain to a system-spanning cascade.

Long term — **The balance**

On average, grains lost at the boundary = grains added at the top. The slope fluctuates around the critical angle. The system self-organizes to this state.

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PART II

The BTW Sandpile Model

Rules, avalanches, and the critical attractor

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BTW Sandpile — Formal Definition

Grid

Square lattice $N \times N$ with **open boundaries** (grains fall off the edge)

State

Height $z(x, y) \in \{0, 1, 2, 3, \ldots\}$ — number of grains at each site

Threshold

Critical height $z_c = 4$ (= number of neighbors in 2D von Neumann neighborhood)

Dynamics

1. DRIVE: Add one grain at a random site: $z(x,y) \to z(x,y) + 1$
2. TOPPLE: While any $z(x,y) \geq 4$:
   $z(x,y) \to z(x,y) - 4$
   $z(\text{neighbors}) \to z(\text{neighbors}) + 1$ (grains at boundary are lost)
3. MEASURE: Record avalanche size s = total number of topplings

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Toppling Rule — Visual

When a site reaches z ≥ 4, it topples: loses 4 grains, each neighbor gains 1.

Before

1	2	0
3	5	1
0	1	2

⟹

After

1	3	0
4	1	2
0	2	2

Notice: The left neighbor (3 → 4) now has z = 4 — it will topple next! This is how avalanches cascade. A single grain can trigger an arbitrarily large chain reaction.

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Why Open Boundaries Matter

Dissipation = Balance

Open boundaries let grains leave the system. In steady state, the average rate of grain loss at boundaries equals the rate of grain addition (one per step). This balance is what makes the critical state an attractor.

Closed Boundaries?

With periodic (closed) boundaries, no dissipation occurs. The total number of grains increases monotonically. The system never reaches a stationary state, and you get no SOC.

The SOC Recipe

Slow driving (adding grains) + Fast relaxation (avalanches) + Dissipation (open boundaries)

→ Separation of timescales: the system is always "just barely stable" when the next grain arrives.
→ Each new perturbation probes the system at its critical state.
→ Power-law distributed response (avalanches of all sizes).

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Toppling as a Linear Operator

The Toppling Matrix

When site (x,y) topples, the height configuration changes by:

$\Delta z(x',y') = -\Delta(x,y)(x',y')$ where Δ is the discrete Laplacian

```
Δ(x,y)(x',y')  =  4   if (x',y') = (x,y)      (central site loses 4)
                 -1   if (x',y') ∈ neighbors   (each neighbor gains 1)
                  0   otherwise
```

Key Property: Abelian

The final configuration after an avalanche does not depend on the order of topplings. This makes the model exactly solvable in many respects (Dhar, 1990). The "Abelian sandpile" is a deep mathematical object related to chip-firing games and algebraic graph theory.

Connection to CA

The sandpile is a CA with a threshold! The update rule $z \to z - 4$ if $z \geq 4$, applied in parallel, is exactly a totalistic cellular automaton. This is why we implement it in NumPy (like Weeks 1–2) rather than Mesa — there are no agents making decisions.

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PART III

Power Laws and Scaling

The signature of criticality

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Power-Law Distributions

Definition

A random variable X follows a power law if:

$$P(X = s) \propto s^{(-\tau)} \qquad \text{or equivalently:} \quad \log P(s) = -\tau \log s + \text{const}$$

Why Power Laws Matter

- No characteristic scale (scale-free)
- Heavy tails: extreme events are rare but not negligible
- Straight line on log-log plot
- Signature of critical phenomena
- Divergent higher moments (if $\tau \leq 3$)

BTW Sandpile Exponents

Avalanche size s	$\tau_s \approx 1.27$
Avalanche area a	$\tau_a \approx 1.37$
Duration t	$\tau_t \approx 1.49$
Radius r	$\tau_r \approx 1.78$

(2D, exact values debated — these are numerical estimates)

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How (Not) to Fit a Power Law

Common Mistakes

1. Linear fit to log-log histogram
Seems natural, but bin sizes distort the distribution. Small bins at large s have huge noise, pulling the fit.

2. Ignoring the lower cutoff
Power laws only hold above some smin. Including small avalanches biases τ.

3. Not testing the hypothesis
Many distributions look "roughly linear" on a log-log plot but aren't power laws (log-normal, stretched exponential, etc.).

The Right Way (Clauset et al. 2009)

1. Use Maximum Likelihood Estimation (MLE) for τ, not least-squares on binned data:

$$\hat{\tau} = 1 + n \left[\Sigma \ln(s_i / s_{min})\right]^{-1}$$

2. Estimate smin using Kolmogorov-Smirnov distance (KS test)

3. Compare with alternative distributions (log-normal, exponential) using likelihood ratio test

4. Use logarithmic binning for visualization (not fitting)

```
pip install powerlaw   # Python implementation
```

SOC in Nature

Earthquakes — $b \approx 1.0$

Magnitude-frequency: $\log_{10} N = a - bM$

Forest fires — $\tau \approx 1.2$

Burn area follows power law (Malamud et al.)

Solar flares — $\tau \approx 1.8$

Peak flux and total energy

Neural avalanches — $\tau \approx 1.5$

Cortical network activity (Beggs & Plenz 2003)

City sizes — $\tau \approx 2.0$

Rank-size distribution of populations

1/f noise — $S(f) \propto 1/f$

Ubiquitous in electronic, biological, financial signals

Caution: not all power laws indicate SOC. Careful statistical testing is essential!

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PART IV

Implementation in NumPy

Vectorized sandpile simulation

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Sandpile Engine — Core

```python
class Sandpile:
    """BTW Abelian sandpile on N×N grid."""
    def __init__(self, N, z_crit=4):
        self.N = N
        self.z_crit = z_crit
        self.grid = np.zeros((N, N), dtype=int)

    def add_grain(self, x=None, y=None):
        if x is None:
            x, y = np.random.randint(self.N, size=2)
        self.grid[x, y] += 1

    def relax(self):
        """Topple until stable. Return avalanche size."""
        size = 0
        while True:
            unstable = self.grid >= self.z_crit
            if not unstable.any():
                break
            size += unstable.sum()
            self.grid[unstable] -= self.z_crit
            # Distribute grains to 4 neighbors
            self.grid[1:, :]  += unstable[:-1, :]
            self.grid[:-1, :] += unstable[1:, :]
            self.grid[:, 1:]  += unstable[:, :-1]
            self.grid[:, :-1] += unstable[:, 1:]
        return size
```

NumPy, not Mesa

No agents here — just a grid with a threshold rule. Same vectorized style as Weeks 1–2. Open boundaries are implicit: grains "falling off" are never added back.

Key Insight

Neighbor distribution via array slicing is the same trick we used for GoL's convolution — but here it's the discrete Laplacian on unstable sites only.

Open boundaries = we simply don't add to the non-existent neighbors.

Running the Sandpile

```python
def run_sandpile(N=64, n_grains=100000, warmup=10000):
    """Run sandpile and collect avalanche statistics."""
    sp = Sandpile(N)
    avalanches = []

    for i in range(n_grains):
        sp.add_grain()
        size = sp.relax()

        # Record only after transient (warmup)
        if i >= warmup:
            avalanches.append(size)

    return sp, np.array(avalanches)
```

```python
sp, avalanches = run_sandpile(N=64, n_grains=100000)
```

```python
# Visualization of the sandpile height field
fig, ax = plt.subplots(figsize=(7, 7))
im = ax.imshow(sp.grid, cmap='YlOrRd',
               interpolation='nearest')
plt.colorbar(im, label='Height z(x,y)')
ax.set_title('Steady-State Height Configuration')
```

Important Details

`warmup`

Discard transient to reach steady state

`N=64`

Good balance of speed vs finite-size effects

`n_grains`

Need ~10^5 for decent statistics

`size=0`

Most grains don't trigger avalanches!

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Analyzing Avalanche Statistics

```python
# Log-binned histogram (the right way to visualize)
def log_binned_histogram(data, n_bins=30):
    """Create logarithmically spaced bins."""
    data_pos = data[data > 0]
    bins = np.logspace(
        np.log10(data_pos.min()),
        np.log10(data_pos.max()),
        n_bins + 1
    )
    counts, edges = np.histogram(data_pos, bins=bins)
    centers = np.sqrt(edges[:-1] * edges[1:])  # geometric mean
    widths = edges[1:] - edges[:-1]
    density = counts / (widths * len(data_pos))
    mask = counts > 0
    return centers[mask], density[mask]

# Plot
centers, density = log_binned_histogram(avalanches)
plt.loglog(centers, density, 'o', color='#4A9ECC')
# MLE fit line (see exercises)
```

Log Binning

For visualization, use logarithmic bins — equal width in log space. This reduces noise at large s where events are rare.

For fitting, always use MLE (not the histogram).

Geometric mean of bin edges gives the correct center on a log scale.

Expected: straight line on log-log with slope $-\tau \approx -1.27$

Finite-size effect: The power law has a cutoff at $s_{max} \propto N^D$ where D is a fractal dimension. Larger grids → wider power-law regime.

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PART V

Analysis and Exercises

Putting SOC to the test

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The Course So Far: Three Paradigms

	CA (Weeks 1–2)	ABM (Week 3)	SOC (Week 4)
Entities	Cells	Agents	Cells (CA with threshold)
Update	Deterministic	Utility-based	Threshold-driven
Boundaries	Periodic	Periodic (torus)	Open (dissipative)
Criticality	Rule-dependent	At T = Tc (tuned)	Self-organized
Key measure	Entropy	Segregation S	Avalanche size P(s)
Signature	Complexity classes	Phase transition	Power laws
Tool	NumPy	Mesa	NumPy

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Week 4 Exercises

Exercise 1 — **Avalanche Size Distribution** — 45 min • Core

Run the sandpile for N = 64, 100000 grains (10000 warmup). Plot the avalanche size distribution with log-binning. Fit the exponent τ using MLE. Compare with the accepted value $\tau \approx 1.27$.

Exercise 2 — **Finite-Size Scaling** — 60 min • Core

Run for N = 16, 32, 64, 128. Plot P(s) for all sizes on the same log-log axes. Show that the cutoff grows with N. Try to collapse the curves using $P(s) = s^{(-\tau)} f(s/N^D)$.

Exercise 3 — **Transient to Criticality** — 45 min • Core

Starting from empty grid, record avalanche sizes over time. Plot a running average. How many grains are needed before the system reaches the critical state? How does this scale with N?

Exercise 4 — **Avalanche Shapes and Duration** — 60 min • Advanced

Modify the code to record the duration (number of toppling rounds) and spatial extent (bounding box) of each avalanche. Plot the duration distribution P(t). Measure the relationship $s \propto t^{\gamma}$ between size and duration.

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Key Takeaways

- SOC = criticality as an attractor, not a parameter value. No tuning needed.
- The BTW sandpile: slow driving + threshold toppling + open boundaries → power-law avalanches
- Power laws $P(s) \propto s^{(-\tau)}$ are the signature of scale-free, critical behavior
- Fitting power laws requires care: use MLE + KS test, not naive log-log regression
- The sandpile is a threshold CA — back to NumPy. Agency (Mesa) not needed here.
- SOC appears across physics, biology, earth science, neuroscience, and finance

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Looking Ahead: From Emergence to Intelligence

Weeks 1–4 Recap

We've built a complete foundation: emergence from local rules (CA), agents with preferences (Schelling/Mesa), criticality without tuning (SOC). Now we add learning.

- **Weeks 5–6: Reinforcement Learning**
  Agents that learn from experience — Q-learning, multi-agent RL
- **Weeks 7–8: Foundation Models**
  Transformers, prompting, tool use — from neurons to LLMs
- **Weeks 9–10: LangChain & LangGraph**
  The Mesa→LangGraph bridge we've been building toward
- **Weeks 11–12: Multi-Agent LLM Systems**
  Full circle: emergence in AI agent systems

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Questions?

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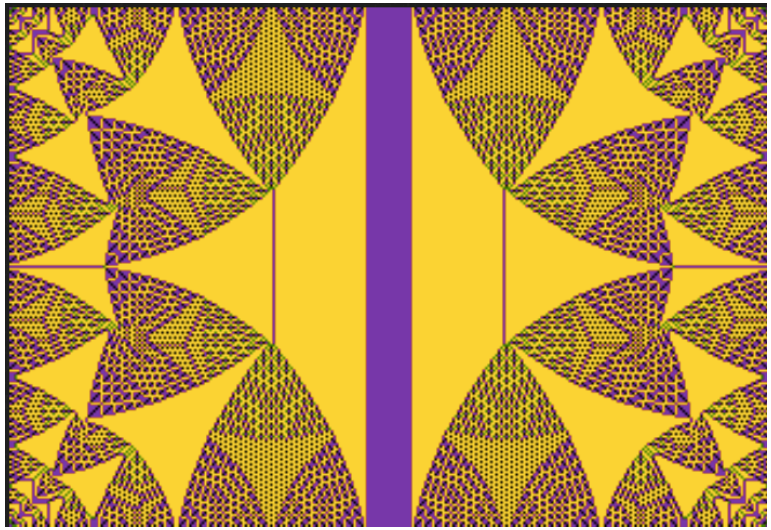
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